

FarmFleet AI

Smart Labour and Farm Machinery Sharing Platform

FarmFleet AI unifies agricultural equipment rental, labour hiring, AI-driven crop planning, weather intelligence, digital payments, PDF reporting and multilingual delivery into a single mobile-first platform for Indian farmers, equipment owners and agricultural labour.

The Problem

▪ Fragmented, informal equipment access	▪ No centralized labour discovery
▪ Limited data-driven crop planning	▪ No weather-linked farming guidance

Rental, advisory, labour and payment channels remain informal, region-limited and digitally disconnected from one another.

The Solution

Discover	→	Plan	→	Rent	→	Hire	→	Pay	→	Monitor	→	Report
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Platform at a Glance

54 FRONTEND ROUTES	20 CONTROLLERS	14 MONGODB MODELS	18 ROUTE FILES	8 LANGUAGES	3 USER ROLES
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Core Capabilities

▪ Equipment Marketplace	▪ AI Crop Planner	▪ Weather Intelligence	▪ Labour Marketplace
▪ Digital Payments	▪ PDF Reports	▪ Notifications	▪ Multilingual UI



FarmFleet AI

How It Works

Equipment Types

For Owners

Reviews

FAQ

English

Continue As

INDIA'S AI-POWERED SMART FARM EQUIPMENT RENTAL PLATFORM

Rent. Share. Grow.

Access verified tractors, harvesters, and farm machinery when you need it — booked in minutes, delivered to your field.

Continue as Equipment Owner

List your equipment and manage bookings

Continue as Renter

Find and book equipment near you

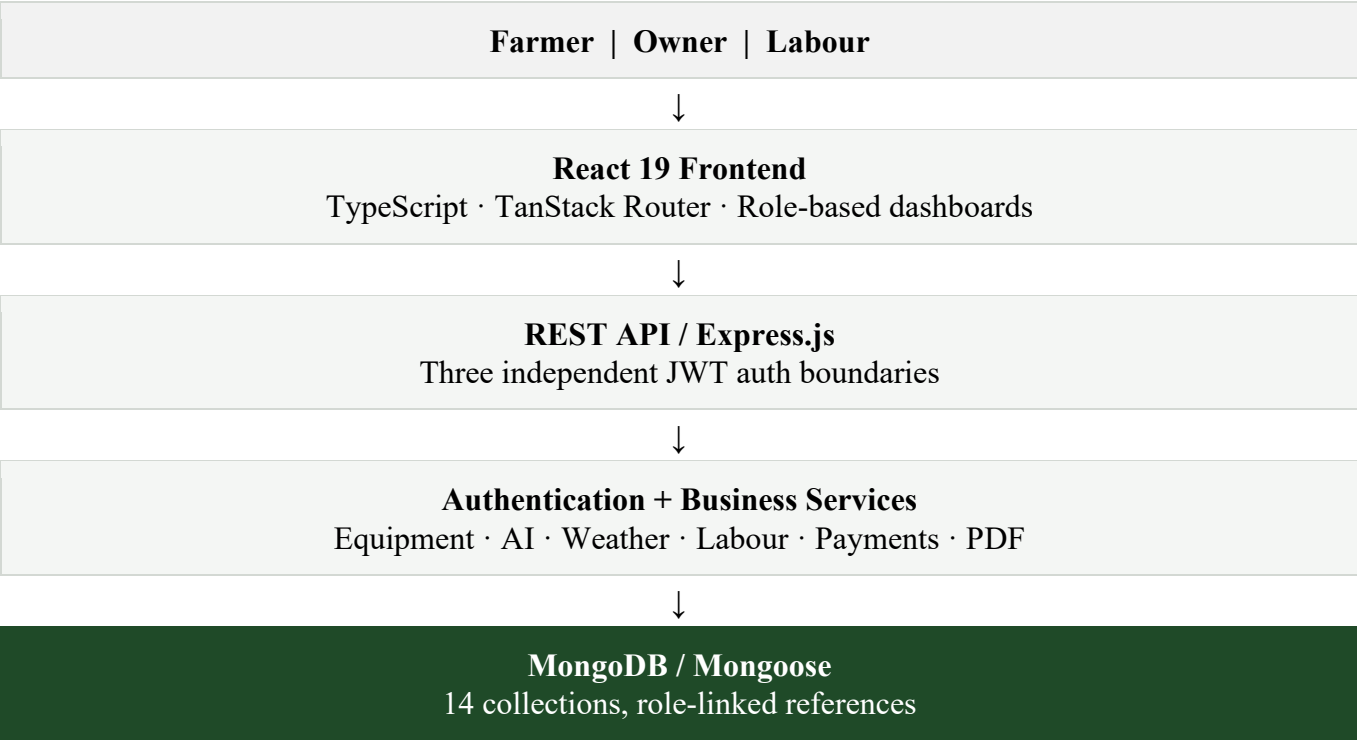
Continue as Labour

Accept farm work and equipment operation requests



1. System Architecture

FarmFleet AI follows a three-tier, role-based architecture: a React single-page frontend, a REST API backend built on Express, and a MongoDB data layer, augmented by five external services for AI, weather, payments, media and email.



External services: Google Gemini · OpenWeatherMap · Razorpay · Cloudinary · Nodemailer

Technology Stack

Layer	Technologies
Frontend	React 19, TypeScript, TanStack Router, TanStack Query, Tailwind CSS v4, Framer Motion, React Hook Form, Zod, Axios, Radix UI, Recharts, i18next
Backend	Node.js, Express.js, MongoDB, Mongoose, JWT, node-cron
AI / Data	Google Gemini, OpenWeatherMap
Payments	Razorpay
Media / Docs	Cloudinary, Multer, PDFKit
Communication	Nodemailer / Gmail SMTP
Build	Vite, @vitejs/plugin-react

2. End-to-End System Workflow

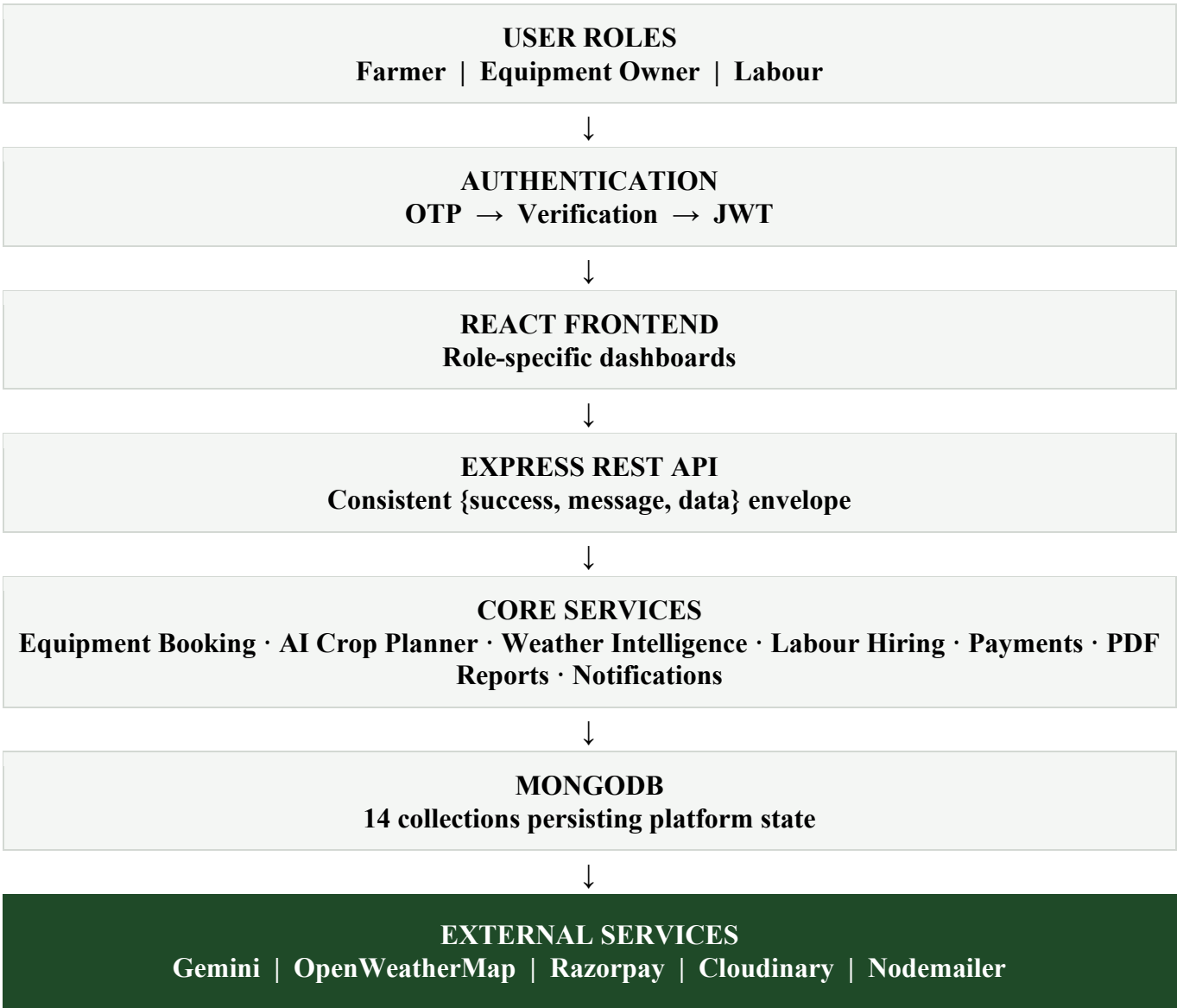
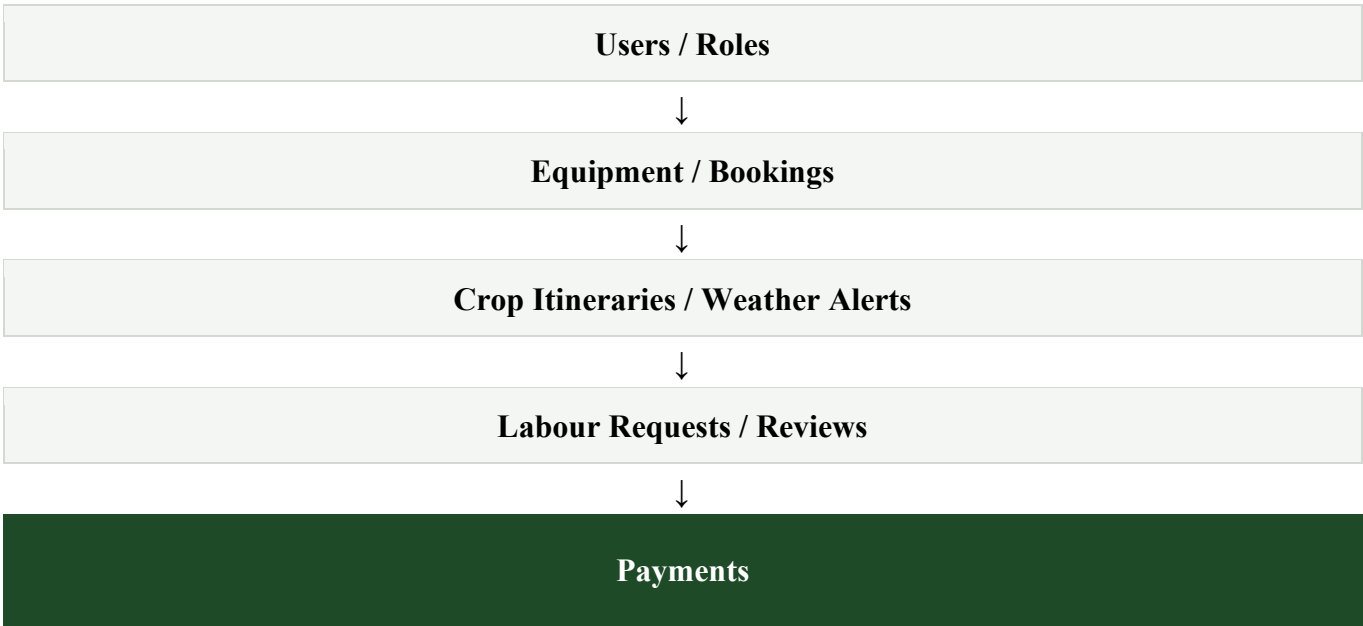


Figure 1. FarmFleet AI end-to-end architecture and service workflow

Technical Observations

- Role-specific authentication isolates access across Farmer, Owner and Labour flows.
- REST APIs connect frontend workflows with backend business services.
- AI and weather processing operate as dedicated, independently retried services.
- MongoDB persists platform state across all role and transaction collections.
- External services provide AI, weather, payments, media and email capabilities.

3. Data & API Design



Core Collections (of 14)

Collection	Purpose
farmers / owners / labours	Role profiles
equipment	Rental inventory
bookings	Equipment bookings
cropitineraries	AI-generated farming plans
payments	Payment transactions
labourrequests	Labour hiring workflow
weatheralerts	Weather-triggered alerts
reviews / labourreviews	Ratings

API Surface

/api/farmer /api/owner /api/labour /api/equipment /api/booking
/api/labour-request /api/payment /api/ai /api/weather
/api/pdf /api/reviews /api/earnings /api/contact

Authentication



Three role-specific middlewares (authMiddleware.js, ownerAuthMiddleware.js, labourAuth.js) verify the bearer token and reject invalid or expired tokens with 401.

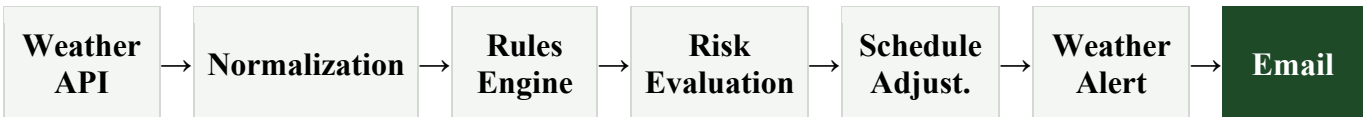
4. Intelligence Layer

AI Crop Planner



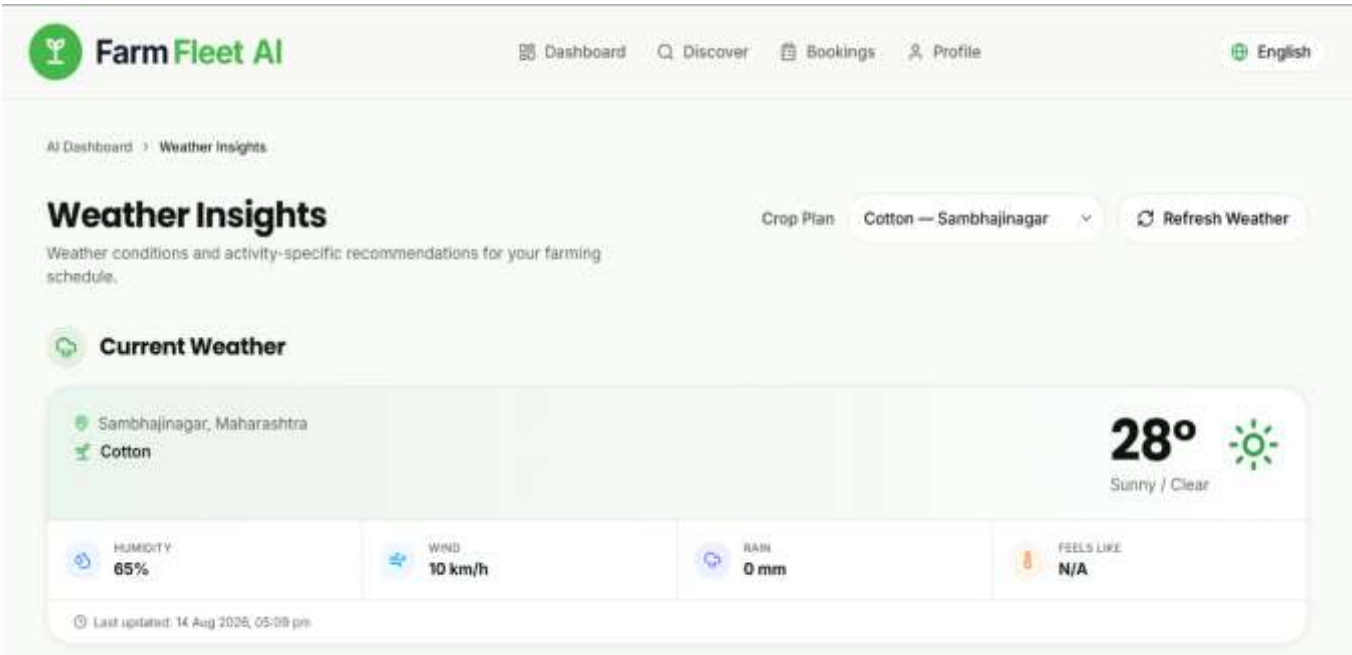
Inputs: crop, state, district, soil type, land area, water source, budget.
promptBuilder.js geminiService.js responseParser.js
responseParser.js normalizes inconsistent or malformed Gemini output into one stable schema before persistence.

Weather Intelligence



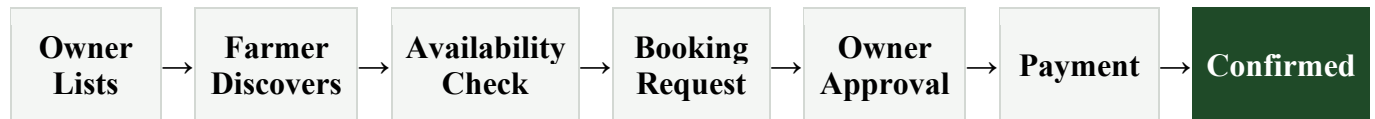
Risk Factor	Threshold
Wind	> 30 km/h
Temperature	> 38°C
Low temperature	< 10°C
Humidity	> 85%
Heavy rain	> 64.5 mm
Moderate rain	> 15.6 mm

930+ lines of weather rules and monitoring logic, automated monitoring at 7:00 AM and 6:00 PM IST.



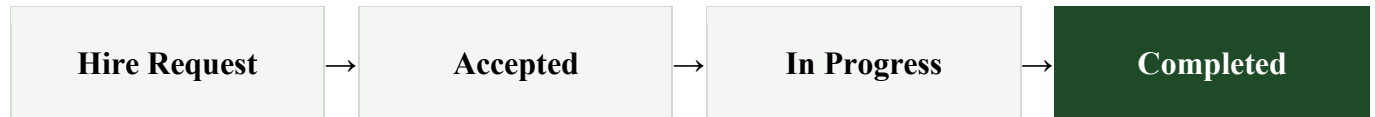
5. Marketplace & Transaction Engine

Equipment Marketplace



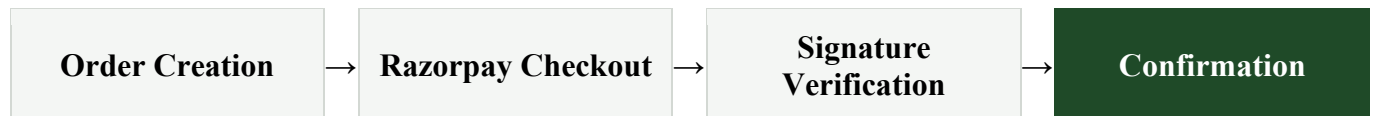
Categories: Tractor · Rotavator · Cultivator · Seed Drill · Sprayer · Harvester · Plough · Thresher · Power Weeder

Labour Marketplace

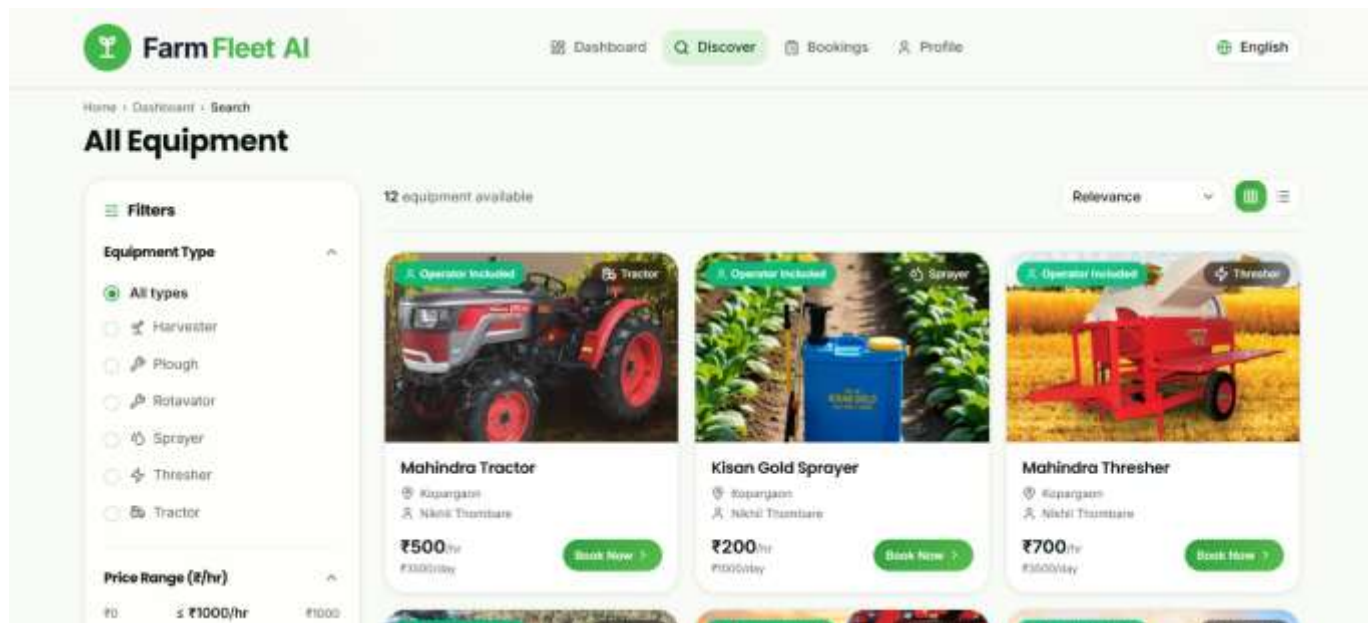


OTP-based work verification confirms job completion before payment release.

Payment Engine



- Duplicate payment prevention via existing-payment checks before order creation.
- HMAC-SHA256 signature recomputed and compared with `crypto.timingSafeEqual()`.
- Server-side webhook fallback verification if the client callback is missed.
- Cash payments separately confirmable by owners or labour.



6. Implementation & Validation

Implementation Scale

54 FRONTEND ROUTES	20 CONTROLLERS	14 MODELS	18 ROUTE FILES	90+ KB WEATHER LOGIC	110+ KB PDF GENERATION
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Key Implementation Components

Component	Implementation
Authentication	OTP + JWT + role-specific middleware
AI Pipeline	Prompt builder + Gemini + response parser
Weather	Rules engine + monitoring + schedule optimizer
Payments	Razorpay + HMAC verification + webhooks
Marketplace	Equipment CRUD + booking lifecycle
Labour	Hire lifecycle + OTP verification
PDF	PDFKit + multilingual templates
Notifications	Nodemailer + weather / itinerary emails

Validation

Test	Expected Behaviour
Expired OTP	Authentication rejected
Malformed AI JSON	Response normalized
Unsafe weather	Schedule delayed + alert
Invalid payment signature	Payment rejected
Booking rejection	Status updated + user notified

Delivered Outputs

AI Crop Plans Localized structured farming itineraries	Weather Monitoring Automated twice-daily schedule safety checks	Digital Reports Downloadable multilingual PDF reports	Integrated Marketplace Equipment + labour + payment lifecycle
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7. Limitations & Future Scope

Current Limitations

- AI recommendations are not agronomist-verified.
- Weather quality depends on OpenWeatherMap coverage and forecast accuracy.
- Real-time AI and weather features require connectivity.
- Offline functionality is currently unavailable.

Future Roadmap

#	Roadmap Item
01	Voice interaction in regional languages
02	AI crop-disease detection
03	IoT sensor integration
04	Drone / satellite data
05	Market-price prediction
06	Government scheme recommendations
07	Offline rural mode
08	Native mobile applications
09	Historical crop-yield analytics

References

APIs

- [Google Gemini API](#)
- [OpenWeatherMap API](#)
- [Razorpay API](#)
- [Cloudinary API](#)

Technical Documentation

- [MongoDB / Mongoose docs](#)
- [Razorpay Integration Guide](#)

Frameworks / Libraries

- React 19, TanStack Router, TanStack Query
- Tailwind CSS, Framer Motion
- Express.js, Mongoose, JWT, node-cron
- PDFKit, Nodemailer, Multer, i18next

Existing Website References

- [JS Farms](#)
- [KisanKamai](#)

FarmFleet AI brings equipment access, labour services, AI-assisted crop planning, weather intelligence, digital payments and multilingual reporting into one integrated agricultural platform.